SAHIL T CHAUDHARY

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SUMMARY

Robotics engineer focused on motion planning, control, and reinforcement learning for humanoid and manipulator systems. CMU MS '25: multi-agent planning and decentralized heterogeneous convoys, trajectory optimization and MPC/MPPI, and PPO robot learning in Isaac Lab; current work on learning-based state estimation and sim-to-real transfer.

EDUCATION

Carnegie Mellon University *Pittsburgh, PA*
Master of Science in Mechanical Engineering – Research — **GPA: 4.0/4.0** *May 2025*

- **Coursework:** Planning and Decision-making, Introduction to Robot Learning, Optimal Control and Reinforcement Learning, Robot Localization and Mapping, Modern Control Theory, Machine Learning, Robot Dynamics and Analysis
- **Scholarship:** Fully funded Master's student
- **Course assistant** for Machine Learning and Artificial Intelligence for Engineers (24787)

Vellore Institute of Technology *Vellore, India*
Bachelor of Technology in Mechanical Engineering — **GPA: 9.05/10.0** *May 2022*

SKILLS

Programming Languages: C++, Python, Bash, MATLAB, Julia

Methods: Reinforcement Learning, Proximal Policy Optimization (PPO), Motion Planning, Trajectory Optimization, Model Predictive Control, Multi-Agent Planning, Decentralized Planning, Kalman Filtering, Sensor Fusion, State Estimation, Sim-to-Real Transfer

Tools: Isaac Lab, Isaac Sim, Gazebo, PyTorch, Gymnasium

Platforms and Software: ROS, ROS 2, PyTorch, DDS, GTSAM, Git, Linux, Docker, SolidWorks, Fusion 360, Ansys

WORK EXPERIENCE

Juxta Technology, Inc. *San Francisco, CA*
Robotics Engineer, AI *September 2025 – Present*

- Developed **learning-based state estimation** using probabilistic and filtering methods for drift mitigation in **neural-inertial navigation and localization** using only IMU, achieving positioning accuracy with ∼**2 m ATE/RTE**
- Built a simulation framework for realistic human-motion trajectories and **synthetic noisy IMU measurements**, incorporating **diffusion** for **sim-to-real transfer**; model trained on synthetic-only data achieves ∼**3 m ATE/RTE** on real-world testing
- Integrated **Kalman Filter**-based estimation with **neural-inertial odometry** and developed heuristic health monitoring and an **LSTM-based critic network** for estimator selection, improving tracking accuracy by up to **33%**
- Combined convolution-based trajectory initialization with **map matching** for global localization, developing hypothesis-pruning and state-selection methods across five candidate locations, achieving **61% success rate**
- Investigated heuristic methods for **yaw-drift correction** and **gyro-bias estimation**, evaluating robustness across trajectories and identifying failure modes that limited generalization

Biorobotics Lab, CMU Robotics Institute *Pittsburgh, PA*
Graduate Research Assistant (Paid during Summer 2024) *August 2023 – May 2025*

- Contributed to the implementation of an **Inverse Kinematics-based optimization** framework for optimal communication node placement
- Spearheaded the development of a **MANET framework** using DDS and ROS to ensure communication fidelity in heterogeneous multi-agent robot convoys, perform network topology repair and recovery behaviors, and enforce a communication boundary
- **Comms-Aware Planning:** Designed a novel **multi-agent planning** algorithm attaining **99% success rate** to maintain communication among robots over radio, by formulating a modified Max-Min Spanning Tree, and validated the algorithm through extensive hardware testing
- **Heterogeneous Convoy Framework:** Developed a decentralized convoy framework integrating RC cars and quadrupeds like Boston Dynamics' Spot, and decentralized planning algorithms enabling intersection rendezvous and coordinated return for exploration missions
- Enhanced the operational efficiency of the **local motion planner by up to 29%**, through waypoint optimization and trajectory smoothing, reducing unnecessary deceleration between waypoints and improving overall robot speed and motion continuity
- **Payload Redesign:** Engineered a modular, serviceable payload for RC cars and quadrupeds, achieving **7% weight reduction** and lower CG while optimizing **sensor FOV and accessibility** across LiDAR, IMU, cameras, compute, and electronics

PUBLICATIONS

Sahil Chaudhary*, Charles Noren*, Burhanuddin Shirose, Bhaskar Vundhurty, Matthew Travers, "Communication Network Construction Behaviors for Robotic Convoying" *Published in GVSETS 2025*

PROJECTS

Blind Manipulation [Website] *Pittsburgh, PA*
Search-Based Planning Lab, CMU RI – Research Project *May 2025 – September 2025*

- Trained a Proximal Policy Optimization (PPO) reinforcement learning agent in **Isaac Lab** for last-mile manipulation, achieving a **94% success rate** in simulation
- Developed a **hierarchical framework** for navigation and manipulation in **unknown environments using only contact sensing**

PinBot – Reinforcement Learning on a Pinball Machine [Website] *Pittsburgh, PA*
Carnegie Mellon University – Course Project *September 2024 – November 2024*

- Trained a Proximal Policy Optimization (PPO) reinforcement learning agent with Gymnasium to play pinball, outperforming a randomized baseline agent by **67%**
- Applying transfer learning to adapt the agent for a physical pinball machine (ongoing work)

Quadruped Path Planner for Dynamic Environments [Website] *Pittsburgh, PA*
Carnegie Mellon University – Course Project *September 2024 – November 2024*

- Demonstrated a global motion planner using C++ and ROS, that accounts for dynamic obstacles and the z-height of the robot
- Executed Lazy-PRM with D-Star Lite and kinodynamic constraints in Gazebo, achieving a **93% success rate**